



Instructor: ..... Date: .....

Learner: ..... Regional Branch: .....

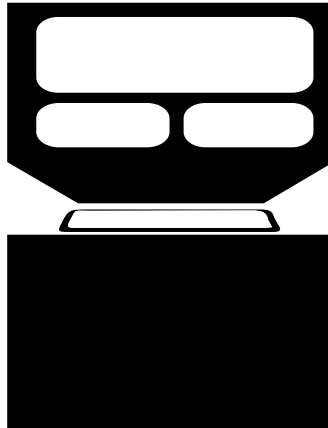
Program: ..... Center: .....

**CONTENT**

- Let's Start!
- Let's Explore!
- Let's Practice!
- Extension Activity
- Target Words
- Learning Points

**Let's Start!**

Back in the day...

**Learning Outcome:**

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.

Give detailed and important information regarding to the history of language programming in different periods of time.

**Ask your partner:**

- What were the functionalities of computers in the past?
- What can we learn from studying the history of language programming?

**Can Do Statements:**

- I can give important information related to the history of software programming and its evolution.
- I can talk about past events related to the history of language programming.
- I describe what Bernoulli numbers are, and their importance for software engineering.

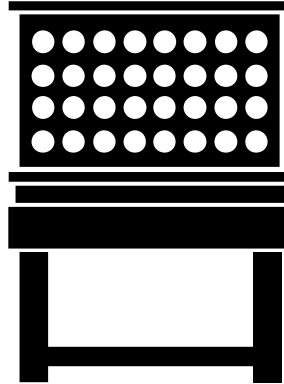




## Let's Explore!

Taking a glimpse\* at the past!

\*To "take a glimpse" is to take a brief look at something.



*Instructions: Read the text below.*

**Frank:** Last night, I watched a documentary about the history of software programming. I have to confess that I want to watch more of these shows, they are really interesting.

**Olivia:** Really? What did you learn?

**Frank:** In 1843, Charles Babbage invented the Analytical Engine, a machine with all the characteristics of a standard computer that could make calculations using Ada Lovelace's first algorithm she created this algorithm when translating the memoir of the Italian mathematician Luigi Manebro. Ada Lovelace, who was a brilliant mathematician of that time, was very interested in Charles Babbage's Analytical Engine; she was the first person to understand the real potential of the machine. She created an algorithm to calculate the sequence of Bernoulli numbers, for this reason, she is considered the first programmer in history.

**Olivia:** Bernoulli numbers?

**Frank:** Yes. A long time ago, Jakob Bernoulli first studied a sequence of rational numbers, that later received his name.

**Olivia:** Wow, what happened next?

**Frank:** Back then, mathematicians started to create computer codes depending on specific applications that people needed. For a period of time, codes that used words and numbers appeared, but mathematicians realized that they could represent them only with numbers. For a while, the evolution of software programming was quite stately. Then, the invention of the Turing machine and the use of punch cards were important breakthroughs for this field that allowed it to reach modernity.

**Olivia:** Is it there, when programming languages appeared?

**Frank:** Indeed! For many years, programming languages had hardware limitations; however, around 1940, the first modern electric computer appeared. Thus, programmers had to develop special assembly language programs that required to do lots of mental efforts. Finally, in 1950 they developed the first programming language that successfully transmitted instructions to computers. The rest is just history...

**Olivia:** Wow! I didn't know you could learn that much from a documentary.

**Frank:** Yeah! It is a shame, though. People spend less time learning about things like these nowadays.





## Let's Practice!

### Some Important Things in Software Programming History

*Instructions: Find the missing words for each sentence.*

#### Word Bank

memoir / limitations / stately / computer / language

1. A document that told Manabre's life. Lovelace translated it to understand a method 1. A document that told Manabre's life. Lovelace translated it to understand a method to calculate Bernoulli numbers.
2. Restrictions produced by the hardware's poor capacity in early computers. Hardware \_\_\_\_\_.
3. The way the evolution of software programming went until the invention of the Turing machine. The first of these electronic devices appeared around 1940.
4. The first of these electronic devices appeared around 1940.
5. In 1950, they developed the first of these programming features, it successfully transmitted instructions to computers. Programming \_\_\_\_\_.



## Let's Practice!

### More History About Language Programming

*Instructions: Proofread the sentences. Write the correct words to complete the sentences.*

1. Some programmers have to create assembly language programs that took great mental efforts to work with them.
2. The evolution of software programming go slowly until the creation of the Turing machine.
3. The Turing machine used punch cards, which was a great breakthrough that yelled to reach the development of modern machines.



## Extension Activity

Let's find out more about programming history.

*Instructions: Complete the following steps. Fill in the Self-Assessment.*

- Look for more important dates related to programming language development.
- Choose 3 dates from what you found.
- Write a paragraph for each date. Include all the facts and people who participated.
- Don't forget to write why it is so important in your opinion.
- Read the paragraphs to a classmate.

### Self-Assessment

Yes

No

- I can look for important information on the internet.
- I can write a paragraph talking about a specific date and the facts that happened in said date.
- I can read out loud the paragraphs I wrote.

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## Target Words

- **memoir** (noun): a biographical document that a person writes about his / her life experiences  
e.g. She created this algorithm when translating the memoir of the Italian mathematician Luigi Manebra.
- **assembly language** (noun): a programming language that permits a programmer command of the machine code to be produced  
e.g. Thus, programmers had to develop special assembly language programs that required lots of mental efforts to do.
- **stately** (adjective): not fast  
e.g. For a while, the evolution of software programming was quite stately.
- **breakthrough** (noun): a great advancement or event in a theoretic field  
e.g. Then, the invention of the Turing machine and the use of punch cards were important breakthroughs for this field that allowed it to reach modernity.
- **hardware limitation** (expression): a restriction produced by the poor capacity of early computers  
e.g. Indeed! For many years, programming languages had hardware limitations; however, around 1940 the first modern electric computer appeared.



## Learning Points

### Simple Past Tense in Affirmative Form, Time Expressions

You use the **Simple Past** to talk about events that started and finished in a specific moment in the past.

Remember that the verbs we use in the **simple past** change depending on their type, regular or irregular. In their past form, the regular verbs always include **-ed, -d** or **-ied** at the end, while the irregular ones **do not follow any changing rule**. Let's see some examples:

**Regular verbs:** *talk - talked, walk - walked, like - liked, study - studied, purchase - purchased*, among others.

**Irregular verbs:** *buy - bought, is/are - was/were, put - put, choose - chose, send - sent*, among others.

As the **Simple Past** tense refers to events in the past, you can emphasize on the moment they took place by using **time expressions**. They tend to be at the beginning of the sentence and we generally put a comma after the expression.

Check the following chart to learn about some of the time expressions we use:

| TIME EXPRESSIONS        |                   |
|-------------------------|-------------------|
| Yesterday               | In 1993 / October |
| Two weeks / days ago    | Last month / year |
| Since last year / month | Lately            |
| Recently                | This morning      |
| For a while             | For some time     |

### Examples

- *Yesterday, we purchased the software for the company.*
- *For some time, the company worked with that software.*
- *A long time ago, Jakob Bernoulli first studied a sequence of rational numbers, that later received his name.*
- *Finally, in 1950 they developed the first programming language that successfully transmitted instructions to computers.*





## Learning Points

### Quantifiers in Simple Present

You use **quantifiers** to mention the quantity or amount of something.

Remember, we use the **Present Simple** to talk about an action that happens right in the present time. Don't forget that we can also use it to talk about generalizations or facts.

Now, we can include **quantifiers** within the **Simple Present** tenses in order to communicate the quantity of an element. Today, we learn about two quantifiers: **More** and **less**.

| Quantifier | Explanation                     |
|------------|---------------------------------|
| More       | A greater quantity of something |
| Less       | A smaller quantity of something |

Remember that we place quantifiers before the noun or element we are talking about.

For example:

- *Day after day, **more** people are interested in learning about the history of modern language programming.*
- *Nowadays, **less** companies are using those types of software.*

### Examples

- *I have to confess that I want to watch more of these programs, they are really interesting.*
- *People spend less time learning about things like these nowadays.*



## Learning Points

### How to Make a Mind Map

A **mind map** is a good strategy we can use to organize, join and classify ideas in a colorful way that allows you to remember important aspects of a topic.

A **mind map** allows you to make a visual organizer of information that you can use at school or for your job in order to retain information in an easy and illustrative way.

You have to go through three simple stages to create a mind map. Let's review them:

#### 1. Planning the Mind Map

This is the most important stage of all. Think about the final objective of the map: Does it explain something? Do you need it to learn something?

Then, think about the main focus of the map, the most important element of all, from which we delimit different characteristics or relationships. Next, brainstorm all the components or concepts that derive from it and that are related to the main concept you chose. If you need so, research or look for extra information on the internet. Think about the relationships and hierarchy that you can extract from the information you got.

#### 2. Creating the Mind Map

Decide the structure that your map will have, spider-like or tree-like. Depending on it, place the most important concept either at the center of the map or at the top. Afterwards, start making relationships that derive from the main concept at the center and join them with lines. Add as much information as needed.

Remember that you can use symbols, drawings and personal conventions that support the relationships and the information you include in the map. Furthermore, remember not to include full sentences or paragraphs, but **keywords and phrases**.

#### 3. Assessing Your Map

Now that you displayed the information on a map, it is time for you to evaluate it. Ask yourself: Is the information clear enough? Start reviewing every piece of information, pay attention to the links you created among them, and assess the pertinence of your drawings or illustrations. If possible, ask somebody you know to read the map in order to get a valuable feedback.

